Anderson Andrei da Silva,

Ph.D in Computer Science

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I am a Postdoctoral Researcher at INRIA Lille, in France, where I study and optimize the energy efficiency of a varied range of applications deployed on serverless computing, such as scientific HPC-AI workflows and RAG Pipelines. Previously, I completed a postdoctoral fellowship at HPE Labs (USA), conducting applied research on improving HPC and AI workflow execution across hybrid HPC-Serverless platforms. I hold a Ph.D. and an M.Sc. in Computer Science from the University of Grenoble Alpes. My Ph.D., was done in collaboration with Ryax Technologies (CIFRE), focused on serverless runtimes for hybrid edge-cloud infrastructures, while my M.Sc. research addressed job scheduling in edge-cloud platforms. I earned my undergraduate degree from the University of São Paulo, Brazil. I have been conducting applied research at the University and Industry focused on the following research topics: **Serverless**, **Cloud and Edge Computing, Scheduling Policies, Energy Efficiency, High Performance Computing, Simulations, Data Science, Computer Networks, Parallel and Distributed Systems**.

EDUCATION

PHD IN COMPUTER SCIENCE, UNIV. GRENOBLE ALPES, GRENOBLE & RYAX TECHNOLOGIES, LYON – FRANCE
January, 2021 – December, 2023.

MASTER DEGREE IN COMPUTER SCIENCE, UNIV. GRENOBLE ALPES, GRENOBLE – FRANCE
September, 2018 – June, 2019. **[FUNDED SCHOLARSHIP]**

UNDERGRADUATE DEGREE IN COMPUTER SCIENCE UNIV. OF SÃO PAULO, SÃO PAULO – BRAZIL
January, 2017 – December, 2020.

PROFESSIONAL EXPERIENCES

POSTDOCTORAL RESEARCH ENGINEER INRIA, LILLE – FRANCE
July, 2025 – Current.

POSTDOCTORAL RESEARCH ENGINEER HEWLETT PACKARD LABS, MILPITAS – UNITED STATES
January, 2024 – December, 2024.

RESEARCH ENGINEER RYAX TECHNOLOGIES, LYON – FRANCE
January, 2021 – December, 2023.

LANGUAGES

PORTUGUESE: Native; **ENGLISH:** Fluent; **FRENCH:** Fluent.

SKILLS

GENERAL: Design of experiments, Statistical Analysis, Communication, Speech/ Presentations, Scientific Writing, Open-source Software Development.

SOFTWARE ENGINEERING: Kubernetes, Knative, OpenWhisk, Pegasus, R, C, Python, OMP, Julia, Bash, Django, SQL, Neo4j, Mong.DB.

TOOLS AND INFRASTRUCTURE: GNU/Linux, Git, Automated Testing, LaTeX, Emacs

COMPLEMENTARY EXPERIENCES

RESEARCHER, INTERN LABORATOIRE D'INFORMATIQUE DE GRENOBLE, GRENOBLE-FRANCE
[Master's Internship]. February, 2019 - Jun, 2019.

NETWORK ADMINISTRATOR INTERNSHIP UNIVERSITY OF SÃO PAULO, SÃO PAULO-BRAZIL
[Undergraduate Internship] April, 2017 - July, 2018.

COMPUTER MAINTENANCE INTERNSHIP UNIVERSITY OF SÃO PAULO, SÃO PAULO-BRAZIL
[Undergraduate Internship] April, 2015 - April, 2017.

EXCHANGE PROGRAM, Chicago - United States INTRAX English School, CHICAGO - UNITED STATES
[English Language Course] Jun, 2013 - July, 2013 **[FUNDED SCHOLARSHIP]**

SOFTWARE CONTRIBUTIONS

HSMFLOW [Inventor]: A workflow management framework for serverless computing; https://github.com/HewlettPackard/wfm-serverless-hpc

WFBENCH [Contributor]: A synthetic workflow generator framework; https://wfcommons.org/

FUNCTIONBENCH [Contributor]: A collection of benchmarks for serverless computing; https://github.com/ddps-lab/serverless-faas-workbench

BATSIM/ SIMGRID [Contributor]: A batch scheduler simulator. https://github.com/oar-team/batsim

EUROPEAN PROJECTS

PHYSICS - Optimized Hybrid Space-Time Service Continuum in FaaS.
Jan 2021 - December 2023

PUBLICATIONS

1. The Environmental Impacts Of High-Performance Computing : A Literature Review, [**Computer Science Review, 2026**];
2. Automated Task Placement for Hybrid HPC-Serverless Workflows, [1st ECHO Workshop at **Supercomputing 2026**];
3. Assessing Power Usage Effectiveness in Serverless Computing Environments, [11th WoSC at **MIDDLEWARE 2025**];
4. Serverless Workflow Benchmarking with HSMFlow and WfBench, [**MIDDLEWARE 2025**];
5. FOA-Energy: A Multi-objective Energy-Aware Scheduling Policy for Serverless-based Edge-Cloud Continuum. [**ACM SAC 2025**];
6. Enabling HPC Scientific Workflows for Serverless, [6th CANOPIE-HPC Workshop at **Supercomputing 2024**];
7. Serverless Computing for Dynamic HPC Workflows, [19th WORKS Workshop at **SuperComputing2024**];
8. Workflows Community Summit 2024: Future Trends and Challenges in Scientific Workflows, [Zenodo];
9. Towards a Multi-objective Scheduling Policy for Serverless-based Edge-Cloud Continuum, [**CCGRID** 2023].
10. Short-Term Ambient Temperature Forecasting for Smart Heaters, [**ISCC 2021**].
11. Evaluating Computation and Data Placements in Edge Infrastructures through a Common Simulator, [**SBAC-PAD** 2020];
12. Real-time Video Transmission in AD-HOC Networks, [**SBRC Workshop 2018**]

TALKS

1. Exploring the Power Usage Effectiveness in Serverless Computing. [**GreenDays** 2026, Sophia, France].
2. Towards Container layer aware Scheduling Policies. [**DevConf.CZ, 2023**, Brno - Czech Republic]. Talk: https://www.youtube.com/watch?v=Rmj5b_w02Lg .
3. Towards an Energy-Aware Multi-objective Scheduling Policy for Server-based Edge-Cloud Continuum. **[COMPAS 2023**, Annecy, France].
4. Scheduling of Functions on Serverless Platforms. [**6th GRD RSD/ASF WINTER SCHOOL ON DISTRIBUTED SYSTEMS & NETWORK**, Rhône-Alpes - France, 2022].

COMPLEMENTARY ACTIVITIES

CONFERENCES PROGRAMME COMMITTEES:

- SUPER COMPUTING 2026 (SC26), Chicago - United States
- 41TH SYMPOSIUM ON APPLIED COMPUTING 2025 (ACM SAC 2026), Greece
- 31ST INTERNATIONAL EUROPEAN CONFERENCE ON PARALLEL AND DISTRIBUTED COMPUTING. (EURO-PAR 2025), Dresden - Germany

CONFERENCES REPRODUCIBLE ARTIFACTS COMMITTEES:

- SUPER COMPUTING 2024 (SC24), Atlanta - United States

CONFERENCES ORGANIZING COMMITTEES:

- NEW CHALLENGES IN SCHEDULING THEORY 2022, Aussois - France
- SEMANA DA COMPUTAÇÃO, 2019 - 2020, São Paulo - Brazil
- SÃO PAULO SCHOOL OF ADVANCED SCIENCE ON SMART CITIES, São Paulo - Brazil

FROM RESEARCH TO REALITY PODCAST [ORGANIZATION/ PRODUCTION TEAM, 2024]
Reference: https://www.youtube.com/@HPE/podcasts

TEACHING EXPERIENCES

TEACHER ASSISTANT (167h),

at UNIVERSITY OF LILLE, LILLE - FRANCE in French:

- Advanced Operating Systems (27h), Master 2, November 2025
- Advanced Distributed Systems 2 (27h), Master 2, September 2025

at UNIVERSITY GRENOBLE ALPES, GRENOBLE - FRANCE in French and English:

- Parallel and Distributed Systems (3h), Master M1, 2022
- Technical Writing and Speaking (9h), Master M1, 2022
- Introduction to Computer Science (14h), L3, 2022
- Introduction to the C Programming Language (8h), L2, 2022
- Database and Applications (30h), L2, 2022
- Design and Operation of Databases (27h), L3, 2021
- Computer Science and Programming(22), L1, 2021

at UNIVERSITY OF SÃO PAULO, SÃO PAULO, BRAZIL, in Portuguese:

- Concurrent and Parallel Programming (22h), L3, 2020
- Introduction to Computer Science to Math. Sciences (30h), L1, 2019
- Programming Topics (27), L1, 2018